

The transfer function of subset-sum landscapes: rigidity of the gap series off centre

(working title — scaffolding, round 88)

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Abstract

[**TODO** written last, after the theorem statements are fixed.] The three things this paper is to establish: (i) that $\text{lm}_A(n)/\deg_A(n) \rightarrow \Gamma(A)$ holds for every fixed target fraction ρ , not only at the centre — so that Γ is an invariant of the landscape and not of one point of it; (ii) that the ratio is a universal functional of the *local logarithmic slope* of the representation count alone, through an explicit transfer function Φ ; and (iii) that the hypothesis the first statement needs is exactly the convergence of the series that gives the second one — with a profile for which it fails.

1 Introduction

[**TODO** full introduction.] Skeleton of the argument, and what each part rests on:

- The companion papers establish $\text{lm}/\deg \rightarrow \Gamma$ *at the centre* $n = \lfloor T/2 \rfloor$: paper 1 reduces it to a flatness hypothesis and paper 2 discharges that hypothesis unconditionally. The obvious question — whether Γ is a property of the landscape or of its centre — is what this paper answers.
- The combinatorial layer (classification, exact stratification, sandwich) was proved *for every target* in paper 1 and is formally verified; nothing there needs redoing. [[Lean-verified](#)]
- The minor-arc estimates of paper 2 bound $|G_B(\theta)|$ as a function of θ only. They do not refer to the extraction point m , which enters $r_B(m) = \frac{1}{2\pi} \int F_B(\theta) e^{-im\theta} d\theta$ only through a phase. They therefore transfer to off-centre targets unchanged. [[quoted from paper 2](#)]
- What is new is the middle: a tilted local limit theorem and a cancellation lemma for the odd orders in λ . [**TODO** [fable-5](#), [spec_t3rigid_r085.md](#)]

2 Notation, and what is inherited

[**TODO** transcribe the definitions from paper 1 in a form that does not assume $n = T/2$.] Fix throughout an increasing sequence $A = (a_1, \dots, a_k)$ of odd positive integers, $T = \sigma(A)$, $S_2 = \sum a_i^2$, $V = S_2/4$, and a target n . Write $\rho = n/T$ and $x = \frac{1}{2} - \rho$. For $d \geq 1$ put

$$N_d = \#\{i : a_i \leq 2d\}, \quad \sigma_d = \sum_{a_i \leq 2d} a_i, \quad s_d = \sum_{a_i \leq 2d} a_i^2, \quad \delta_d = d + \frac{\sigma_d}{2},$$

and let $B_d = \{a_i : a_i > 2d\}$ be the layer- d ensemble, of variance $V_d = (S_2 - s_d)/4 = V - s_d/4$.

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Definition 2.1 (local logarithmic slope). $\lambda(n)$ is the logarithmic derivative of the representation count at the target,

$$\lambda(n) = -\frac{d}{dm} \log r_A(m) \Big|_{m=n} = \frac{1}{4} \log \frac{r_A(n-2)}{r_A(n+2)} + O(\cdot).$$

Remark 2.2 (why the slope and not the offset). The Gaussian approximation $\lambda \approx (T/2 - n)/V$ is what one writes down first, and it is wrong at the 10% level for the purposes of this paper. **[TODO write out: the substitution turns a 15–18% drift in the measured coefficient into 0.03%, and it is the reason an earlier draft of this programme reported a spurious non-integer exponent.]** This is not a cosmetic choice: every statement below is false at the stated precision if $(T/2 - n)/V$ is substituted for λ .

Definition 2.3 (the tilt). P_s is the product measure on subsets under which a_i is present with probability $p_a = (1 + e^{sa})^{-1}$, and $s = s(n) > 0$ is chosen so that $\sum_a a p_a = n$; thus P_s is the exponential tilt of the uniform measure whose mean sits at the target.

Remark 2.4 (the tilt and the slope agree to first order). s and λ are the same quantity to leading order but not exactly: s makes the tilted *mean* equal n , whereas λ is the slope of $\log r_A$ at n , and the two differ by the mean–mode gap. Measured on three profiles,

$$\frac{s(n)}{\lambda(n)} = 1 + \frac{c_A}{k} + O(k^{-2}), \quad c_A \approx 1.85 \text{ (odds)}, 2.03 \text{ (primes)}, 2.87 \text{ (squares)},$$

stable in x and clean in k over $k = 100, \dots, 220$. **[experimentally confirmed; tilt_r088]** The statements below are formulated in λ , and *not* in s : substituting s multiplies the residual of Table 1 by three, exactly as the λ^2 leading behaviour of Φ predicts. **[experimentally confirmed; sres_r088]**

3 Rigidity

Theorem 3.1 (T3-Rigid, to be proved). **[TODO statement from spec_t3rigid_r085.md.]** *Informally: for the odd primes and every fixed $\rho \in (0, 1)$, $\text{lm}_P(\lfloor \rho T \rfloor) / \deg_P(\lfloor \rho T \rfloor) \rightarrow \Gamma(P)$, under the hypothesis of §6.*

[experimentally confirmed] The evidence the theorem is modelled on. With $\text{ratio}(\rho, k) = \lfloor \text{lm} / \deg \rfloor / \Gamma(P_k)$, the spread over ρ contracts monotonically:

k	40	52	64	76	88	100
$\max_\rho - \min_\rho$	0.01051	0.00283	0.00257	0.00162	0.00113	0.00083
$\max_\rho \text{ratio} - 1 $	0.00921	0.00276	0.00244	0.00156	0.00109	0.00081

over $\rho \in [0.20, 0.50]$, by exact integer computation. **[TODO full table; state the granularity guard $\deg \geq 10^4$ and the two correctness checks (agreement with brute force over all 2^k subsets; the reflection identity $\text{lm}(n) = \text{lm}(T - n)$, which is exact).]**

4 The transfer function

Theorem 4.1 (correction theorem, to be proved). **[TODO statement.]** *Informally:*

$$\frac{\text{lm}_A(n)}{\deg_A(n)} = \Phi(\lambda(n)) (1 + o(1)), \quad \Phi(\lambda) = 1 + \sum_{d \geq 1} 2^{1-N_d} e^{-\lambda^2 s_d / 8} \cosh(\lambda \delta_d). \quad (1)$$

Three features of (1) deserve to be stated before any proof.

Remark 4.2 (the ratio sees the target only through one number). Φ depends on A alone. The target enters only through $\lambda(n)$. This is a stronger statement than rigidity: rigidity says the limit does not move, (1) says how it fails to move.

Remark 4.3 ($\lambda = 0$ recovers paper 1). The layer weight $e^{-\lambda^2 s_d/8}$ is 1 at $\lambda = 0$, so $\Phi(0) = 1 + \sum_{d \geq 1} 2^{1-N_d}$, which is the window identity $W_D(A) = \Gamma(A) + (2D + 1)2^{-k}$ of paper 1 — already formally verified. Numerically, $\Phi(0) - \Gamma(A)$ is 0, $-4.7 \cdot 10^{-14}$ and $-2.7 \cdot 10^{-15}$ for the three sequences of Table 1.

Remark 4.4 (each layer is a smaller ensemble). The factor $e^{-\lambda^2 s_d/8}$ is not a fudge: it is what remains when the Gaussian normalisations of B_d and of A fail to cancel, because $V_d = V - s_d/4$. Expanding to second order gives the coefficient of λ^2 as $\delta_d^2 - s_d/4$ rather than δ_d^2 . [TODO write out the two lines; note that the correction is termwise smaller than the term it corrects, $s_d/4 \leq d\sigma_d/2 < \delta_d^2$, so the sum stays positive.]

4.1 Verification

[experimentally confirmed] Table 1 evaluates (1) at the measured $\lambda(n)$ against the exactly computed lm/deg .

$x = \frac{1}{2} - \rho$	0.06	0.08	0.10	0.15	0.20	0.25	0.30
odds, $k = 220$	0.84%	0.85%	0.85%	0.86%	0.88%	0.92%	0.97%
squares, $k = 170$	1.66%	1.67%	1.68%	1.70%	1.74%	1.80%	1.89%
primes, $k = 220$	0.91%	0.92%	0.92%	0.94%	0.96%	0.99%	1.05%

Table 1: Residual $|\text{measured} - \Phi(\lambda)|$ as a fraction of the centred signal $\text{lm}/\text{deg} - [\text{lm}/\text{deg}]_{n=T/2}$. Absolute residuals are 10^{-5} to 10^{-6} ; quoting them as significant figures of the ratio would be meaningless, since the signal itself lives in the fifth and sixth digit.

Remark 4.5 (why the residual is quoted this way). Against the *uncentred* signal the same residuals read 2% to 300%, the large values occurring precisely where the signal is smallest. Neither number is wrong; the centred one is the one the theorem is about, because $\Phi(0) = \Gamma$ is an identity and the finite- k offset $[\text{lm}/\text{deg}]_{n=T/2} - \Gamma$ is a separate quantity of size 10^{-6} . [TODO keep this remark: an earlier draft of the programme reported “agreement to five or six significant figures” for a model that was 12% wrong on the signal.]

Remark 4.6 (the residual obeys a $1/k$ law). Table 1 is one column of a k -scan. Writing $R_A(k)$ for the residual in the sense of the caption, the measurement over $k = 60, \dots, 220$ gives

$$k R_A(k) \longrightarrow \begin{cases} 1.85\% & \text{odds } (k = 60, 100, 140, 180, 220 : 2.29, 2.00, 1.91, 1.87, 1.85), \\ 2.01\% & \text{primes } (k = 100, \dots, 220 : 2.38, 2.06, 2.03, 2.01), \\ 2.82\% & \text{squares } (k = 130, \dots, 190 : 3.78, 2.85, 2.83, 2.82). \end{cases}$$

Two things follow. First, the residual is $O(1/k)$ and therefore vanishes, which is the form Theorem 4.1 asserts; the table’s 1% is a finite- k number and not a floor. Second, the squares row is the loosest of the three because c_A is larger, not because the decay is different: squares obey the same law with a constant 1.52 times that of the odd numbers. The constants c_A coincide numerically with those of Remark 2.4 — the same mean–mode gap seen twice — but substituting s for λ does not remove the residual, so the coincidence is a shared order of magnitude and not a shared cause. [experimentally confirmed; tilt_r088, sres_r088] [TODO fable: is c_A the second Edgeworth coefficient? The odd orders cancel by the reflection symmetry, so the first survivor is of exactly this size.]

5 Universality of the rate function

[experimentally confirmed, $k = 16..56$, three reference sequences] [TODO full table.] With $\hat{I}_A(\rho) = -k^{-1} \log(\deg_A(\lfloor \rho T \rfloor)/2^k)$ and $D_k = \hat{I}_P - \hat{I}_{\text{ref}}$, the difference tends to 0 for each of three reference sequences: the odd numbers, the arithmetic progression $6n + 5$, and a random odd sequence of the same range. The sign is a property of the reference, not a universal fact — it is positive against the odd numbers, negative against the random sequence, and changes sign near $k = 48$ against the progression.

Remark 5.1 (the primes sit near the random sequence). At $k = 56$ the distance to a random odd sequence is 0.0016 against 0.0146 to the arithmetic progression — an order of magnitude. [TODO develop: this is the direct evidence for the reading that the main term is universal and the arithmetic of the sequence appears only in the correction.]

6 A profile for which the hypothesis fails

The hypothesis Theorem 3.1 needs is the convergence, with room, of the series in (1) — qualitatively, that $N_d \rightarrow \infty$ fast enough. [TODO state it precisely once fable-5’s skeleton fixes the form.]

[experimentally confirmed] It is not vacuous. For the odd-ified cubes $a_i = 2\lfloor i^3/2 \rfloor + 1$:

profile	odds	primes	squares	cubes
N_d at $d = 10$ (any k)	10	7	4	2
$Q(0) = \Gamma^{-1} \sum 2^{-N_d} (\delta_d^2 - s_d/4)$	20.3	50.4	916	381 904

Only $\{1, 9\}$ ever lie below 20, at any k , so 2^{-N_d} never damps the growing δ_d^2 and the series turns over only near $d \approx 10^4$. At $k = 70$ the finite-size offset is still $1.75 \cdot 10^{-2}$ while the signal is $1.7 \cdot 10^{-11}$: nine orders of magnitude, so the profile is not merely slow but undecidable at any accessible size. [TODO present this as the counterexample that makes the hypothesis non-vacuous, not as a failure.]

7 Honest scope

[TODO in the style of paper 1 §10 and paper 2 §10.] The classification to fill in:

- *Formally verified*: the combinatorial layer, inherited from paper 1, for every target.
- *Proved*: [TODO after fable-5’s skeleton.] The minor-arc input is quoted from paper 2 and is m -independent; paper 2’s own ineffectivity through the Siegel–Walfisz constant is inherited.
- *Experimentally confirmed, with the range stated*: rigidity (§3), the transfer function to 0.8–1.9% of the signal (Table 1), universality (§5), the profile-dependent constant $4(2\alpha + 1)/(\alpha + 1)$ verified on four profiles.
- *Not claimed*: the residual 0.8–1.9% of Table 1 is consistent with finite size and shrinks in k , but we do not claim it is zero.

Remark 7.1 (approaches that did not work). [TODO as in paper 2, keep the dead ends with their reasons.] At least: the tilted-measure (Γ_x) reading of off-centre targets, which the data rejected — rewriting the collapse in the exact tilt parameter makes it worse, not better; and the reading of the correction as a third-cumulant effect, which was an artefact of using the Gaussian slope.

Data availability

[**TODO** scripts and logs, as in papers 1 and 2.]

References

- [1] K. Amauchi, *The gap series of an integer sequence: an arithmetic invariant governing subset-sum landscapes*. [**TODO** arXiv number]
- [2] K. Amauchi, *Asymptotic flatness of subset-sum landscapes of primes: the sub-peak spectrum and the constant $\sqrt{3}/2$* . [**TODO** arXiv number]